

Network Theory

Course Introduction

22.07.2025

Course code: BAECE103

L T P C – 3 1 0 4

Faculty: *Dr. Elizabeth Rufus*

Professor (Higher Academic Grade)

School of Electronics Engineering

Faculty Room: TT341

Mobile:967734996

Open hours: *Wednesday: 15:00 to 16:00 hrs*

Thursday : 15:00 to 16:00 hrs

Course Objectives

1. To familiarize circuit laws and theorems and to solve DC and AC circuits.
2. To understand mathematical methods such as the Laplace transform, Fourier transform and graph theory for solving circuit problems.
3. To develop an ability to analyze two-port networks, first and second-order circuits.

Course Outcomes

Will be able to:

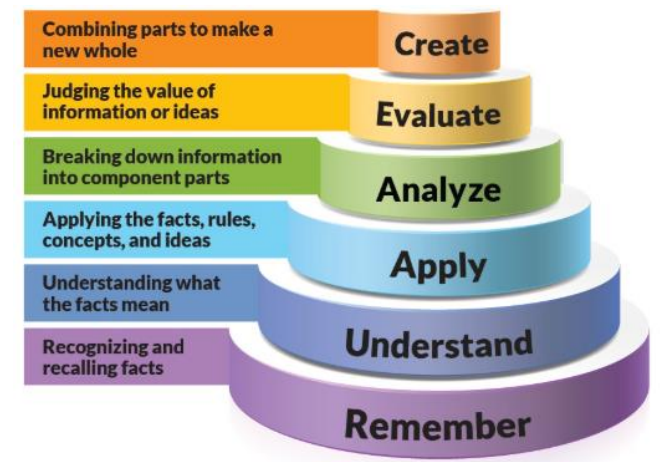
1. Compute DC circuit parameters using circuit laws and theorems.
2. Determine the natural and step responses of RC, RL, and RLC circuits.
3. Apply appropriate techniques and theorems for sinusoidal steady-state analysis.
4. Apply the transforms to circuits for steady-state and transient analysis.
5. Compute two-port network parameters and network graph matrices for circuit analysis.

Books

1. Charles K. Alexander, Matthew N. O. Sadiku, **Fundamentals of Electric Circuits**, 2020, 7th Edition, McGraw-Hill Higher Education, United States.
2. Abhijit Chakrabarti, **Circuit Theory Analysis and Synthesis**, 2018, 7th Revised Edition, Dhanpat Rai & Co., India.
3. A. Sudhakar, Shyammohan S. Palli, **Circuits and Networks: Analysis and Synthesis**, 2017, 5th Edition, McGraw-Hill, India.
4. M. E. Van Valkenburg, T.S. Rathore, **Network Analysis**, 2019, Revised 3rd Edition, Pearson Education, England.
5. Mahmood Nahvi, Joseph A. Edminister, **Schaum's Outline of Electric Circuits**, 2018, 7th Edition, McGraw-Hill Education, United States.
6. W.H.Hayt, J.E.Kemmerly & S.M.Durbin, **Engineering Circuit Analysis**, 2019, 9th Edition, McGraw-Hill Higher Education, United States.

Assessment

*Bloom's Taxonomy Levels: **2-Understand** and **3 - Apply***



Continuous Assessment 1 (CAT1) – Module 1 and Module 2 ----- 15 marks (CO1, CO2)

Continuous Assessment 2 (CAT2) – Module 3 and Module 4 ----- 15 marks (CO3, CO4)

Final Assessment Test (FAT) – Modules, 1, 2, 3, 4, 5 ----- 40 marks (CO1, 2, 3, 4, 5)

Quiz – 13.08.25 ----- 10 marks (CO1, CO2)

Class project I – 18.09.25 (deadline to upload) ----- 10 marks (CO3, CO4)

Class project II – 04.11.25 (deadline to upload) ----- 10 marks (CO1, 2, 3, 4, 5)

Total marks ----- 100 marks

All the Best

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God Bless